

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Data Structures Practical

Practical - 25

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Class : SYIT	Batch : 01
Date of Assignment : 01-09-26	Date/Time of Submission :01-09-26

Q.25 Handle missing values in a dataset. Replace missing values with the column mean.

Code:

```
object desk extends App {

  val sales:List[java.lang.Double] = List(100.0, 200.0, null, 300.0, null)

  val v = sales.filter(_ != null)

  val mean = v.map(_.toDouble).sum / v.length

  println("Mean Sales: " + mean)

  val n = sales.map { value =>
    if (value == null) mean
    else value
  }

  println("Original Data:")
  println(sales.mkString(", "))

  println("After Replacing Missing Values:")
  println(n.mkString(", "))
  println("Himanshu S037")
}
```

Output:

```
Mean Sales: 200.0
Original Data:
100.0, 200.0, null, 300.0, null
After Replacing Missing Values:
100.0, 200.0, 200.0, 300.0, 200.0
Himanshu S037
```

Practical 26.

Perform logistic regression using Breeze. Classify a dataset with binary labels.

Code:

```
import breeze.linalg._

object desk extends App {

  val x = DenseVector(1.0, 2.0, 3.0, 4.0, 5.0) // Dataset

  val y = DenseVector(0.0, 0.0, 0.0, 1.0, 1.0) // Binary labels

  // Initial weight and bias
  var weight = 0.0
  var bias = 0.0

  // Sigmoid function
  def sigmoid(z: Double): Double = {
    1.0 / (1.0 + math.exp(-z))
  }

  // Train the model
  for (i <- 1 to 1000) {
    for (j <- 0 until x.length) {
      val prediction = sigmoid(weight * x(j) + bias)
      val error = prediction - y(j)
      weight = weight - 0.01 * error * x(j)
      bias = bias - 0.01 * error
    }
  }
}
```

```

    }

}

// Classify the data
println("Predictions:")
for (i <- 0 until x.length) {
    val probability = sigmoid(weight * x(i) + bias)
    val result =
        if (probability >= 0.5) 1 else 0
    println("Input = " + x(i) + ", Probability = " + probability + ", Class = " + result)
}
}

```

Output:

```

C:\Users\mvt01\.jdk\ms-17.0.20\bin\java.exe ...
Predictions:
Input = 1.0, Probability = 0.0515501057576029, Class = 0
Himanshu S037
Input = 2.0, Probability = 0.16412826954644827, Class = 0
Himanshu S037
Input = 3.0, Probability = 0.4149888799701166, Class = 0
Himanshu S037
Input = 4.0, Probability = 0.7193154698480789, Class = 1
Himanshu S037
Input = 5.0, Probability = 0.9025176083649726, Class = 1
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```